

Training Uzi

PPO for lane-dominant 1v1 Honor of Kings agents

Rank #3 | 131 / 67 | 66.2%

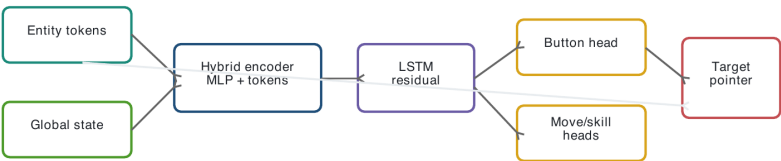
Uzi-style policy: pressure, spacing, target selection, tower conversion

METHOD

Make semantics agree

- Structured tokens for heroes, towers, minions, bullets, cakes, and global state.
- Button-conditioned target pointer: target logits depend on the selected action button.
- Action-mask repair removes normal attacks with only None/Self targets.
- Rewards emphasize safe trades, last hits, tower pressure, danger avoidance, and small action-quality shaping.

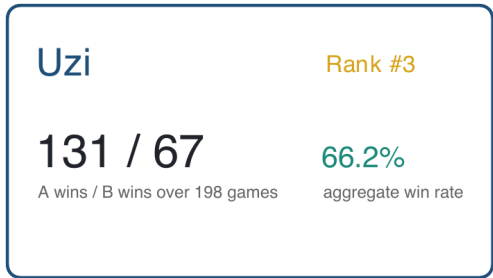
Uzi PPO architecture



Target logits are conditioned on the selected button, aligning actions with entity slots.

RESULTS

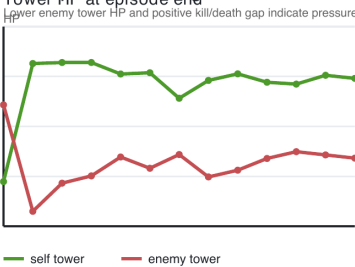
Strong external finish



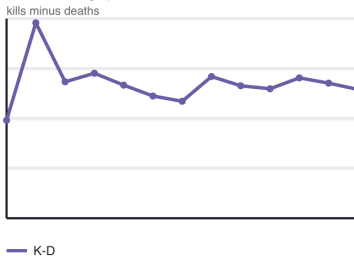
PPO training progression from local logs



Lane pressure diagnostics



Kill-death gap



CASE STUDIES

Honest failures explain design

- Target semantics: real probes showed Soldier slots align by runtime id after nearest-four selection.
- Recall: reward shaping and exploration could start Button9, but channel completion stayed too sparse; final run disables recall.
- Gongsun Li: Arli mark features are observable, but shared lane rewards underfit hero-specific timing.

Recall was a real negative result

Buttons action count

Buttons was exposed in several experiments, but final config disables recall.

Synthetic recall reward

recall_recover

Synthetic frame probes across code history

Cells summarize deterministic code behavior, not full match evaluation.

version	mask	soldier	recall	Arli mark
current working tree	blocks	10,20,30	avail	nonzero
ppo diy feature swap	blocks	-	avail	nonzero
hybrid encoder	blocks	-	avail	nonzero
ppo owned features r.	blocks	10,20,30	avail	nonzero
training observabili.	blocks	10,20,30	avail	nonzero
multi hero curriculum	blocks	10,20,30	avail	nonzero
recall reward shaping	blocks	10,20,30	avail	nonzero
temporary recall exp.	blocks	10,20,30	avail	nonzero
recall button legal	blocks	10,20,30	avail	nonzero
recall ignition moni.	blocks	10,20,30	avail	nonzero
recall channel state	blocks	10,20,30	avail	nonzero